

|   | Caso                                                         | Intervalo de Confianza                                                                              | Contraste de Hipótesis                                                                                                                                                                          | Estadístico del Contraste                                           | Región de rechazo                                                                                                                                            |
|---|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Para $\mu$ de $N(\mu, \sigma^2)$ ,<br>$\sigma^2$ conocida    | $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$                                                  | $H_0 : \mu = \mu_0; H_1 : \mu \neq \mu_0$<br>$H_0 : \mu \leq \mu_0; H_1 : \mu > \mu_0$<br>$H_0 : \mu \geq \mu_0; H_1 : \mu < \mu_0$                                                             | $z_c = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$         | $\{ z_c  > z_{\alpha/2}\}$<br>$\{z_c > z_\alpha\}$<br>$\{z_c < -z_\alpha\}$                                                                                  |
| 2 | Para $\mu$ de $N(\mu, \sigma^2)$ ,<br>$\sigma^2$ desconocida | $\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$                                                  | $H_0 : \mu = \mu_0; H_1 : \mu \neq \mu_0$<br>$H_0 : \mu \leq \mu_0; H_1 : \mu > \mu_0$<br>$H_0 : \mu \geq \mu_0; H_1 : \mu < \mu_0$                                                             | $t_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$             | $\{ t_c  > t_{n-1, \alpha/2}\}$<br>$\{t_c > t_{n-1, \alpha}\}$<br>$\{t_c < -t_{n-1, \alpha}\}$                                                               |
| 3 | Para $\mu$ caso general,<br>$n$ grande                       | $\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$                                                       | $H_0 : \mu = \mu_0; H_1 : \mu \neq \mu_0$<br>$H_0 : \mu \leq \mu_0; H_1 : \mu > \mu_0$<br>$H_0 : \mu \geq \mu_0; H_1 : \mu < \mu_0$                                                             | $z_c = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim N(0,1)$              | $\{ z_c  > z_{\alpha/2}\}$<br>$\{z_c > z_\alpha\}$<br>$\{z_c < -z_\alpha\}$                                                                                  |
| 4 | Para una proporción,<br>$p$ , caso general<br>( $n$ grande)  | $\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$                                      | $H_0 : p = p_0; H_1 : p \neq p_0$<br>$H_0 : p \leq p_0; H_1 : p > p_0$<br>$H_0 : p \geq p_0; H_1 : p < p_0$                                                                                     | $z_c = \frac{(\hat{p} - p_0)}{\sqrt{\hat{p}\hat{q}/n}} \sim N(0,1)$ | $\{ z_c  > z_{\alpha/2}\}$<br>$\{z_c > z_\alpha\}$<br>$\{z_c < -z_\alpha\}$                                                                                  |
| 5 | Para $\sigma^2$ de $N(\mu, \sigma^2)$                        | $\left( \frac{(n-1)s^2}{\chi_{n-1, \alpha/2}^2}, \frac{(n-1)s^2}{\chi_{n-1, 1-\alpha/2}^2} \right)$ | $H_0 : \sigma^2 = \sigma_0^2; H_1 : \sigma^2 \neq \sigma_0^2$<br>$H_0 : \sigma^2 \leq \sigma_0^2; H_1 : \sigma^2 > \sigma_0^2$<br>$H_0 : \sigma^2 \geq \sigma_0^2; H_1 : \sigma^2 < \sigma_0^2$ | $\chi_c^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi_{n-1}^2$          | $\{\chi_c^2 \notin [\chi_{n-1, 1-\alpha/2}^2, \chi_{n-1, \alpha/2}^2]\}$<br>$\{\chi_c^2 > \chi_{n-1, \alpha}^2\}$<br>$\{\chi_c^2 < \chi_{n-1, 1-\alpha}^2\}$ |

Nota:  $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

$f = \frac{(s_1^2 / n_1 + s_2^2 / n_2)^2}{(s_1^2 / n_1)^2 / (n_1 - 1) + (s_2^2 / n_2)^2 / (n_2 - 1)}$  Se elige el entero más próximo a esta cantidad.

|    | Caso                                                                                                                                          | Intervalo de Confianza                                                                                            | Contraste de Hipótesis                                                                                                                                                                                      | Estadístico del Contraste                                                                                          | Región de rechazo                                                                                                                                             |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | Para $\mu_1 - \mu_2$ de $N(\mu_1, \sigma_1^2)$ y $N(\mu_2, \sigma_2^2)$ $\sigma_1^2, \sigma_2^2$ conocidas                                    | $(\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$                                     | $H_0 : \mu_1 - \mu_2 = d_0; H_1 : \mu_1 - \mu_2 \neq d_0$<br>$H_0 : \mu_1 - \mu_2 \leq d_0; H_1 : \mu_1 - \mu_2 > d_0$<br>$H_0 : \mu_1 - \mu_2 \geq d_0; H_1 : \mu_1 - \mu_2 < d_0$                         | $z_c = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0,1)$                   | $\{ z_c  > z_{\alpha/2}\}$<br>$\{z_c > z_\alpha\}$<br>$\{z_c < -z_\alpha\}$                                                                                   |
| 7  | Para $\mu_1 - \mu_2$ de $N(\mu_1, \sigma_1^2)$ y $N(\mu_2, \sigma_2^2)$ , $\sigma_1^2, \sigma_2^2$ desconocidas, $\sigma_1^2 \neq \sigma_2^2$ | $(\bar{x} - \bar{y}) \pm t_{(n_1+n_2-2), \alpha/2} s_p \sqrt{1/n_1 + 1/n_2}$                                      | $H_0 : \mu_1 - \mu_2 = d_0; H_1 : \mu_1 - \mu_2 \neq d_0$<br>$H_0 : \mu_1 - \mu_2 \leq d_0; H_1 : \mu_1 - \mu_2 > d_0$<br>$H_0 : \mu_1 - \mu_2 \geq d_0; H_1 : \mu_1 - \mu_2 < d_0$                         | $t_c = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \cdot \sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2}$                    | $\{ t_c  > t_{n_1+n_2-2, \alpha/2}\}$<br>$\{t_c > t_{n_1+n_2-2, \alpha}\}$<br>$\{t_c < -t_{n_1+n_2-2, \alpha}\}$                                              |
| 8  | Para $\mu_1 - \mu_2$ de $N(\mu_1, \sigma_1^2)$ y $N(\mu_2, \sigma_2^2)$ , $\sigma_1^2, \sigma_2^2$ desconocidas, $\sigma_1^2 = \sigma_2^2$    | $(\bar{x} - \bar{y}) \pm t_{f, \alpha/2} \sqrt{s_1^2/n_1 + s_2^2/n_2}$                                            | $H_0 : \mu_1 - \mu_2 = d_0; H_1 : \mu_1 - \mu_2 \neq d_0$<br>$H_0 : \mu_1 - \mu_2 \leq d_0; H_1 : \mu_1 - \mu_2 > d_0$<br>$H_0 : \mu_1 - \mu_2 \geq d_0; H_1 : \mu_1 - \mu_2 < d_0$                         | $t_c = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \sim t_f$                                | $\{ t_c  > t_{f, \alpha/2}\}$<br>$\{t_c > t_{f, \alpha}\}$<br>$\{t_c < -t_{f, \alpha}\}$                                                                      |
| 9  | Para $\mu_1 - \mu_2$ caso general ( $n_1$ y $n_2$ grandes)                                                                                    | $(\bar{x} - \bar{y}) \pm z_{\alpha/2} \sqrt{s_1^2/n_1 + s_2^2/n_2}$                                               | $H_0 : \mu_1 - \mu_2 = d_0; H_1 : \mu_1 - \mu_2 \neq d_0$<br>$H_0 : \mu_1 - \mu_2 \leq d_0; H_1 : \mu_1 - \mu_2 > d_0$<br>$H_0 : \mu_1 - \mu_2 \geq d_0; H_1 : \mu_1 - \mu_2 < d_0$                         | $z_c = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \sim N(0,1)$                             | $\{ z_c  > z_{\alpha/2}\}$<br>$\{z_c > z_\alpha\}$<br>$\{z_c < -z_\alpha\}$                                                                                   |
| 10 | Para $p_1 - p_2$ caso general ( $n_1$ y $n_2$ grandes)                                                                                        | $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\hat{p}_1 \hat{q}_1/n_1 + \hat{p}_2 \hat{q}_2/n_2}$               | $H_0 : p_1 - p_2 = p_0; H_1 : p_1 - p_2 \neq p_0$<br>$H_0 : p_1 - p_2 \leq p_0; H_1 : p_1 - p_2 > p_0$<br>$H_0 : p_1 - p_2 \geq p_0; H_1 : p_1 - p_2 < p_0$                                                 | $z_c = \frac{(\hat{p}_1 - \hat{p}_2) - p_0}{\sqrt{\hat{p}_1 \hat{q}_1/n_1 + \hat{p}_2 \hat{q}_2/n_2}} \sim N(0,1)$ | $\{ z_c  > z_{\alpha/2}\}$<br>$\{z_c > z_\alpha\}$<br>$\{z_c < -z_\alpha\}$                                                                                   |
| 11 | Para $\sigma_1^2 / \sigma_2^2$ de $N(\mu_1, \sigma_1^2)$ y $N(\mu_2, \sigma_2^2)$                                                             | $\left( \frac{s_1^2 / s_2^2}{F_{n_1-1, n_2-1, \alpha/2}}, \frac{s_1^2}{s_2^2} F_{n_2-1, n_1-1, \alpha/2} \right)$ | $H_0 : \sigma_1^2 = \sigma_2^2; H_1 : \sigma_1^2 \neq \sigma_2^2$<br>$H_0 : \sigma_1^2 \leq \sigma_2^2; H_1 : \sigma_1^2 > \sigma_2^2$<br>$H_0 : \sigma_1^2 \geq \sigma_2^2; H_1 : \sigma_1^2 < \sigma_2^2$ | $F_c = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1}$                                                                  | $\{F_c \notin [F_{n_1-1, n_2-1, 1-\alpha/2}, F_{n_1-1, n_2-1, \alpha/2}]\}$<br>$\{F_c > F_{n_1-1, n_2-1, \alpha}\}$<br>$\{F_c < F_{n_1-1, n_2-1, 1-\alpha}\}$ |